

Cloud Webserver Security Report

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1. Understanding Customer Challenges and Requirements

1.1 Client Challenges

Current Setup and Issues

The client currently runs a local webserver inside a Docker container.

- Apache is installed, but only default modules are enabled (`authn_`, `authz_`).
- No HTTPS is enforced, and some accounts lacked proper password policies.
- The system has duplicate UIDs and a permissive umask of 022.

Limitations

Scalability

Local-only Docker limits the ability to handle high traffic. No automatic scaling or load balancing is in place.

Compliance

Missing password policies, weak hashing rounds, and absent HTTPS affect regulatory compliance (CIS, industry standards).

Disaster Recovery

No backup strategy or failover mechanism; if the container or host fails, services go offline.

1.2 Audit Results

Baseline Security Posture

Lynis hardening index: 54/100

Critical Vulnerabilities Identified

- Duplicate UIDs (AUTH-9208)
- Missing password hashing rounds (AUTH-9230)
- No PAM password complexity module (AUTH-9262)
- Accounts without expiration dates (AUTH-9282)
- Minimum password age not enforced (AUTH-9286)
- Permissive UMASK 022 (AUTH-9328)

Connection to Client Challenges

- Weak authentication and missing account policies increase security risks.
 - Duplicate UIDs create audit and accountability issues, impacting compliance.
 - Permissive file permissions can expose sensitive data, affecting both security and compliance.
 - Local-only container without failover impacts scalability and disaster recovery.
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2. Evaluating Cloud Service Models

2.1 Service Model Comparison

Client Responsibility

Relevant Vulnerabilities

OS, apps, configs

All vulnerabilities remain the client's responsibility

Apps and configs only

OS-level issues may be managed by provider

Only app usage

Most server-level vulnerabilities handled by provider

2.2 Recommended Service Model

Chosen Model: IaaS

Justification

- Provides control over OS, Apache configs, and security hardening
- Aligns with lab goals to apply custom hardening actions

- Supports scalability via cloud instance resizing, load balancing, and backups

3. Exploring Deployment Options

3.1 Deployment Choices

Security

Disaster Recovery

Moderate, requires strong
hardening

High, provider support

High (internal
control)

Moderate

High

High

3.2 Recommended Deployment Option

Chosen Option: Public Cloud

Justification

- Provides scalable compute for increasing traffic
- Managed networking and backups meet DR objectives
- Lab hardening measures can be applied for compliance

4. Developing a Cloud Solution Recommendation

4.1 Hardening Actions

Issue

Action

Result

Duplicate
UIDs
(AUTH-
9208)

`sudo userdel -r
AnotherUser`

Eliminated
duplicate UID,
improving
accountability

Password
hashing
rounds
(AUTH-
9230)

`Edited /etc/login.defs:
SHA_CRYPT_MIN_R
OUNDS 5000`

Increased
password
strength

Missing
PAM
module
(AUTH-
9262)

`sudo apt install
libpam-passwdqc`

Enforced
password
complexity

No
expiration
(AUTH-
9282)

`sudo chage -E
2026-12-31
<username>`

Credential
rotation

No min
password
age
(AUTH-
9286)

`Set PASS_MIN_DAYS
in /etc/login.defs`

Prevents
reuse

UMASK
022
(AUTH-
9328)

`Set UMASK 027`

Restricted file
permissions

4.2 Cloud Solution Recommendation

Proposed Solution

Deploy the webserver on public cloud IaaS with:

- Hardened Apache container
- Secure user/account policies
- HTTPS enforced
- Backups and monitoring enabled

Justification

Public cloud ensures scalability, high availability, and disaster recovery. Hardening actions improve security posture and compliance.

5. Disaster Recovery Planning

5.1 Proposed Recovery Plan

Backups

- Daily snapshots of container volumes
- Backup `/etc/apache2`
- Store in secure cloud storage

Failover

- Multiple instances behind load balancer
- Auto-scaling groups

Monitoring

- Apache access/error logs
 - CloudWatch or Prometheus alerts
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5.2 Connection to Hardening

- UMASK 027 secures backups
 - Password policies protect recovery access
 - HTTPS ensures secure failover traffic
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6. Reflection on the Cloud Process

6.1 Understanding Challenges

Identifying scalability, compliance, and DR risks guided vulnerability prioritization.

6.2 Evaluating Service Models

IaaS provides control; PaaS/SaaS shift responsibility.

6.3 Deployment Options

Public cloud offers scalability and recovery advantages.

6.4 Hardening

Lynis audit improved security and readiness.

6.5 Disaster Recovery

Hardening supports fast, secure recovery.

6.6 Overall Reflection

Technical hardening and strategic planning work together to deliver secure, scalable cloud systems.

